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NQR and NMR of ^{139}La in Antiferromagnetic $La_2CuO_{4-\delta}$

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Rather complex Nuclear Quadrupole Resonance (NQR) spectrum of ^{139}La in $La_2CuO_{4-\delta}$ observed at low temperatures is successfully interpreted as a signal from a single La site which suffers a combined effect of a quadrupole coupling (coupling constant $\nu_Q=6.38\pm0.02$ MHz with asymmetry parameter $\eta=0.01\pm0.01$) and a small Zeeman term where the internal field ($H_N=0.997\pm0.010$ kOe) is nearly perpendicular to the direction of the maximum field gradient. The spin structure of Cu at low temperatures is suggested to be a simple antiferromagnet with sublattice moments parallel to the a-axis. The temperature dependences of ν_Q and H_N have been measured in detail. The fractional decrease of H_N which is proportional to the Cu sublattice moments is found to be in general accord with the prediction of spin fluctuations theory for weak itinerant antiferromagnets.

§1. Introduction

Since the discovery of high-T_c superconductivity in the La-Ba-Cu-O system by Bednorz and Müller,[1-3] so much effort has been devoted to investigate the structural, chemical and physical properties of the related systems in order to clarify the still-unknown mechanism of the high-T_c superconductivity. The undoped compound La_2CuO_4 is tetragonal at high temperatures but changes into orthorhombic phase at 533 K.[4] The structure at room temperature is slightly distorted K_2NiF_4-type layered perovskite structure as shown in Fig. 1.[5,6] A minimum in the resistivity for a single crystal of La_2CuO_4 was observed at 537 K[7] and the distortion is suggested to be associated with a formation of charge density wave (CDW) that opens an energy gap at the Fermi surface.[6,8] There is also an anomaly in the static magnetic susceptibility.[9,24] The concentration dependence of the anomaly observed in $(La_{1-x}Ba_x)CuO_{4-\delta}$[10] has been interpreted as a spin density wave (SDW) formation for a quasi-two-dimensional tight binding electrons.[11] Thus it seems very important to clarify the electronic ground state of the pure La_2CuO_4 itself since it brings a high-T_c superconductor with a slight modification of the electronic structure. The Nuclear Quadrupole Resonance (NQR) experiment has already started to study the electronic structure from the microscopic view point.[12] However, the reported spectrum of La_2CuO_4 in a low frequency range at low temperature has a complicated structure which consists of four doublets and a single peak.[12] They could not be simply identified as a NQR spectrum of ^{139}La with $I=7/2$ and at least two different La sites have been forced to be assumed unless signals from ^{63}Cu and ^{65}Cu are overlapped.[12] We have also made an extensive NQR and NMR studies of the undoped $La_2CuO_{4-\delta}$, and this paper deals with the interpretation and discussions on the static part of the experimental results for the ^{139}La nuclei.

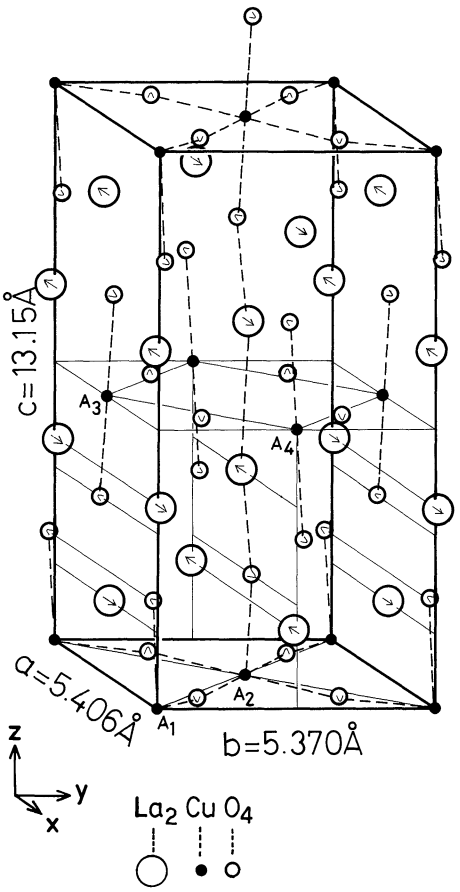


Fig. 1. The crystal structure of La_2CuO_4 (space group: Abma) from ref. 5. Arrows shown in the La and O sites indicate the directions of their displacements.

The temperature dependence of the NQR spectrum and the high field NMR spectrum are presented in §2. We show in §3 that all signals are generated from a single ^{139}La site which is in a large electric field gradient that is slightly deviated from the axial symmetry and subjected to an effect of a weak internal field below $T_N=250$ K. The field is found to be nearly perpendicular to the axis of maximum field gradient. A probable spin structure of Cu in the antiferromagntic state based on the NQR analysis and the calculation of dipole fields are described in §4. The temperature dependence of the quadrupole coupling and the internal field at the ^{139}La site is discussed in §5. Concluding remarks are given in §6.

§2. Experimental Procedure and Results

The sample of $La_2CuO_{4-\delta}$ used in this experiment was prepared by the coprecipitation method (so-called the wet method) from solutions of La- and Cu-nitrates with oxalic acid in appropriate cation ratios with a subsequent heat treatment at 400°C (4 hours), 850°C (12 hours) and 1000°C (10 hours) in air. The sample was checked by powder X-ray diffraction to be the single phase with orthorhombic lattice parameters, $a=5.404$ Å, $b=5.356$ Å and $c=13.155$ Å which agree well with reported values.[5] The temperature dependence of the static susceptibility of the sample is shown in Fig. 2. A peak in the static susceptibility is observed at 242 ± 5 K. A small Curie term seen at low temperatures is probably due to a small amount of paramagnetic impurities ($\sim 0.15\%$ of total Cu atoms if we assume impurities with $S=1/2$).

The zero-field NQR and high-field NMR experiments have been performed by the phase-coherent type pulsed method in the temperature range between 1.3 K and 293 K. To obtain the frequency spectrum, the amplitude of spin-echo signal was examined by point by

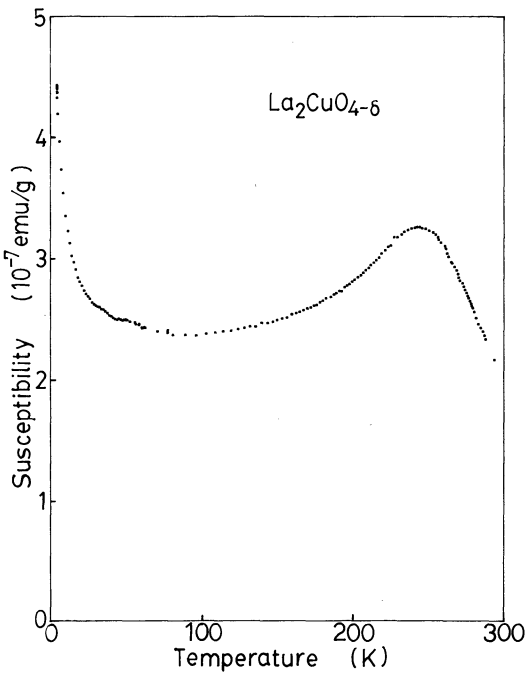


Fig. 2. Temperature dependence of magnetic susceptibility of $La_2CuO_{4-\delta}$ in a static field of 11.8 kOe.

point of frequency of typically every 5 kHz. An intense care has been taken to determine the peak frequency of the spectrum around 19 MHz by observing the zero-cross point of the imaginary part of the signal in the frequency axis. The Fast Fourie Transform (FFT) technique was also employed to examine the spectrum at high temperatures. The accuracy of the obtained peak frequency is estimated to be typically ± 2 kHz.

2.1 *NQR frequency spectrum in the antiferromagnetic state*

The NQR spin-echo frequency spectrum taken at 1.3 K is shown in Fig. 3(a). The peak positions of the spectrum 19.26, 19.02, 12.81, 12.58, 8.08, 7.73, 5.70, 5.35 and 2.38 MHz are found to be nearly the same as those reported.[12] However, the intensities of the doublet at 8.08/7.73 MHz and a single peak at 2.38 MHz are weaker for the present sample than those for a sample prepared by the usual method (so-called the dry method) of simple mixing La_2O_3 with CuO, although experimental intensity in zero-field NQR has only semi-quantitative significance. This will be discussed later.

2.2 *Zero-field NQR frequency spectrum in the paramagnetic state*

The NQR spin echo frequency spectrum at 293 K taken by point by point of frequency of every 5 kHz is shown in Fig. 4. Poor signal-to-noise ratio (S/N=2~3) at room temperature has forced us to average spin-echo signals by 512 times or more to get accurate spectrum. The spectrum in Fig. 4 is a simple NQR spectrum for a single La site with approximately axially symmetric field gradient. The peak frequencies associated with the $\pm 7/2 \leftrightarrow \pm 5/2$, $\pm 5/2 \leftrightarrow \pm 3/2$ and $\pm 3/2 \leftrightarrow \pm 1/2$ transitions were determined to be 18.411 ± 0.002, 12.270 ± 0.002 and 6.142 ± 0.02 MHz, respectively. The increasing line width with going to higher transitions clearly demonstrates that there exists a sizable distribution in the field gradient at the La sites which may be due to the oxygen deficiency in this sample.

2.3 *Temperature dependence of the zero-field NQR spectrum of the doublet of the highest frequency side*

The temperature dependence of the spectrum with double peaks in the highest frequency side at 19.26/19.02 MHz in Fig. 3(a) was studied in detail, and the peak frequencies are depicted in Fig. 5 as a function of temperature. The frequency separation decreases with increasing temperature to vanish at around 250 K where the peak in the static susceptibility has been observed. We associate this temperature to the Néel temperature (T_N) of our sample.

2.4 *NMR field spectrum of ^{139}La in a swept magnetic field*

A typical NMR spin-echo field spectrum of ^{139}La at 4.2 K in a swept magnetic field up to 80 kOe in a superconducting solenoid is shown in Fig. 6 for an operating frequency of 38.00005 MHz. It shows a characteristic quadrupole-split NMR powder pattern for $I=7/2$ with a small internal field.

§3. Interpretation of NQR and NMR Spectrum of ^{139}La

3.1 *NQR spectrum*

The NQR spectrum of ^{139}La is described by a nuclear spin Hamiltonian which is a sum of quadrupolar and Zeeman terms with usual notations,[13]

$$\mathcal{H}=\frac{3e^2qQ}{4I(2I-1)}\left\{I_Z^2-\frac{1}{3}I(I+1)+\frac{1}{6}\eta(I_+^2+I_-^2)\right\}-\gamma_N\hbar\boldsymbol{H}\cdot\boldsymbol{I}, \quad (1)$$

where $I=7/2$, $Q=0.21$(b), $\gamma_N/2\pi=0.60144$ (MHz/kOe) for ^{139}La. The asymmetry parameter η for the electric field gradient (EFG) tensor is defined by

$$\eta=(V_{XX}-V_{YY})/V_{ZZ}, \quad (2)$$

where

$$|V_{ZZ}| \geq |V_{YY}| \geq |V_{XX}|, \quad (3)$$

is assumed in the principal axis system of EFG. A schematic energy level diagram of the spin system obtained by diagonalizing the Hamiltonian (eq. (1)) is shown in Fig. 7. The

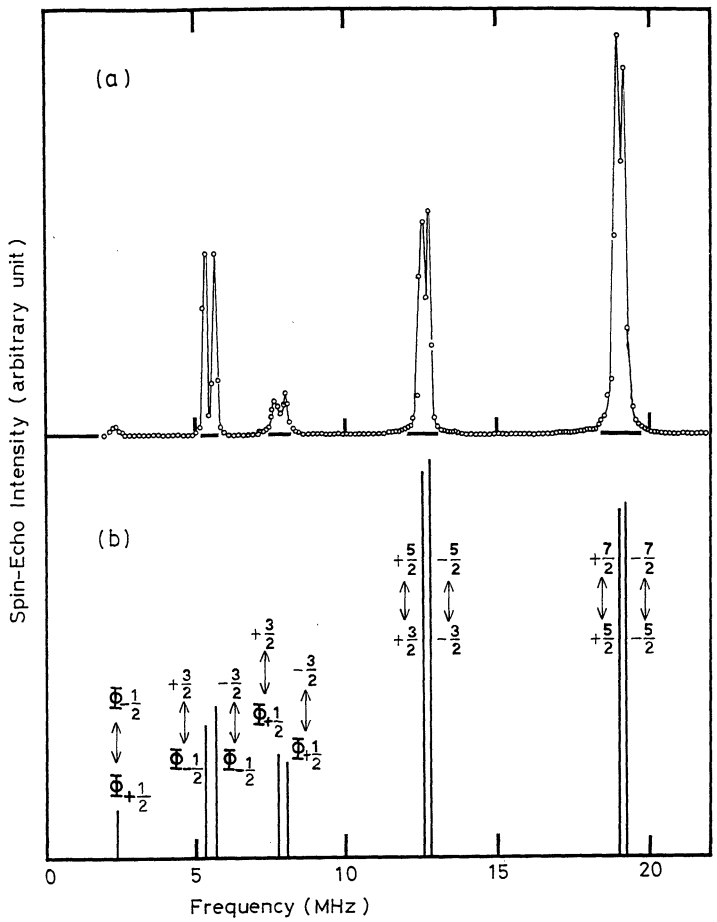


Fig. 3. (a) A spin-echo spectrum of ^{139}La at 1.3 K in zero applied field for a sample of $La_2CuO_{4-\delta}$ prepared by the coprecipitation method. (b) A calculated NQR spectrum of ^{139}La with a set of parameters: $\nu_Q=6.38$ (MHz), $\eta=0.01$, $H_N=0.997$ (kOe), $\theta=78°$ and $\phi=0°$ (see text). A calculated intensity for each transition is shown with the length of each bar.

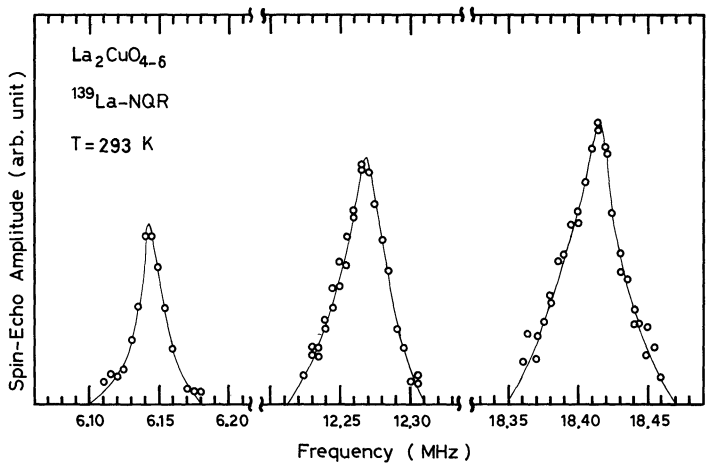


Fig. 4. A spin-echo spectrum of ^{139}La NQR at 293 K in zero applied field.

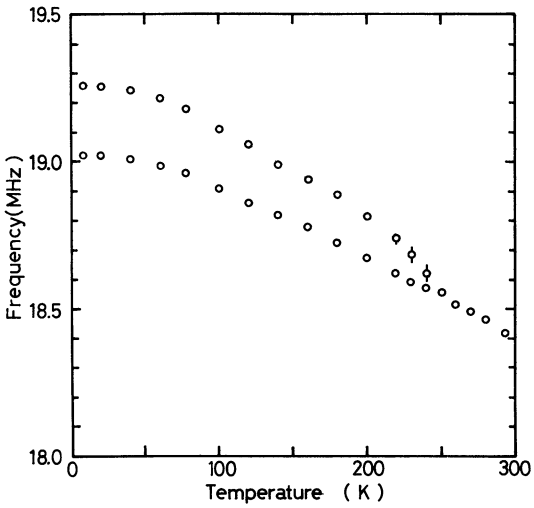


Fig. 5. Temperature dependence of the resonance frequencies of the highest double peaks of ^{139}La NQR in $La_2CuO_{4-\delta}$.

spectrum in Fig. 4 at 293 K can be simply interpreted as a NQR spectrum from a single La site for a case of non-vanishing η in Fig. 7. The obtained parameters for ν_Q and η are as follows:

$$\nu_Q = 3e^2qQ/2I(2I-1)h = 6.13 \pm 0.01 \text{ (MHz)}, \quad (4a)$$

and

$$\eta = 0.01 \pm 0.01. \quad (4b)$$

However, the spectrum in Fig. 3(a) at 1.3 K cannot be explained from a single La site if the magnetic field is absent. Even if one assumes two or more sites of La, it is easily seen that the spectrum in Fig. 3(a) cannot be understood from so-called "PQR" energy levels. Since SDW ground state is suggested in pure La_2CuO_4,[10,11] and the spectrum at 293 K is simply interpreted by a single La site, we try to interpret the spectrum by assuming a non-zero internal field at a single ^{139}La site. Energy levels as are shown on the right hand side in Fig. 7 are calculated numerically by solving a secular equation for 8×8 matrix representation of the Hamiltonian (eq. (1)) for various sets of parameters of η, H, θ and ϕ, where θ and ϕ are polar angles of $\boldsymbol{H}$ with respect to the

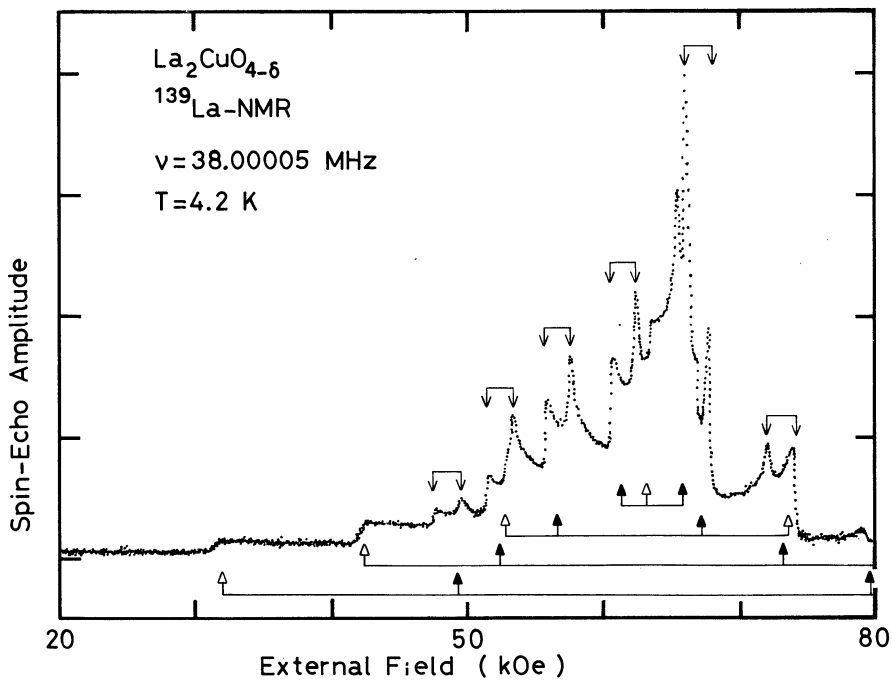


Fig. 6. A spin-echo spectrum of ^{139}La NMR in $La_2CuO_{4-\delta}$ at 38 MHz in a swept magnetic field. Singular points in the powder spectrum at $\theta=0$ and 90° with zero internal field are shown with white and black arrows on the bottom, respectively ($\nu_Q=6.38$ MHz, η is neglected). Splitting of about 1.9 kOe shown with pairs of arrows on the top are due to the internal field of 0.997 kOe at the ^{139}La site.

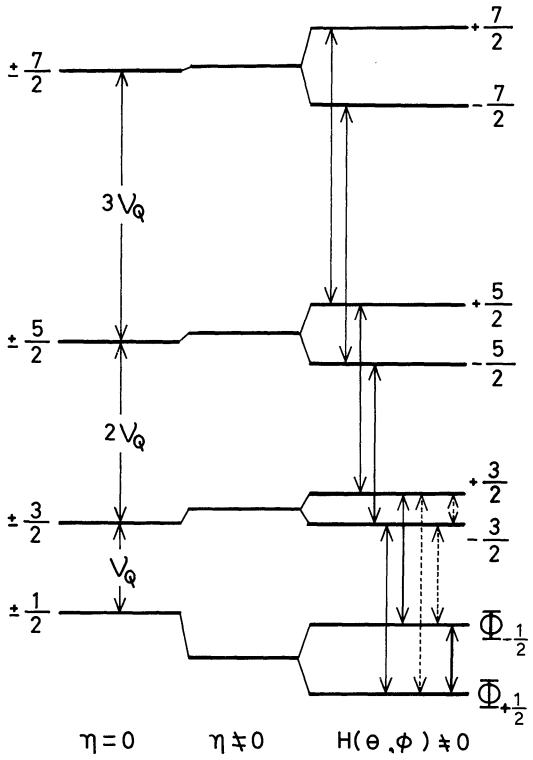


Fig. 7. A schematic energy level diagram of ^{139}La ($I=7/2$). The case of axially symmetric field gradient ($\eta=0$), a case with non-vanishing η and a general case with nonvanishing magnetic field are shown. Allowed transitions and "forbidden" transitions are shown with solid and broken lines, respectively.

principal axes of EFG.

Examples of calculated resonance frequencies for EFG somewhat deviated from the axial symmetry ($\eta=0.1$) are shown in Figs. 8(a)–8(c) as a function of reduced field strength $f_r=\gamma_N H/2\pi\nu_Q$ for the direction of the magnetic field parallel to the three principal axes. In the case where the field H is parallel to the Z-axis of EFG, the effect of the field is just to split the $\pm m\leftrightarrow\pm(m-1)$ transition into two with a separation approximately given by

$$\Delta\nu\simeq(\gamma_N/2\pi)2H. \tag{5}$$

Therefore the spectrum with doublets appears for very weak field, but the number of doublets is only three. This does not accord with the experimental data in Fig. 3(a). However, situation becomes quite different when the magnetic field H is perpendicular to the Z-axis as shown in Figs. 8(b) and 8(c) for examples. The Zeeman term lifts the degeneracy of the originally degenerated $|\pm1/2\rangle$ NQR levels and a drastic change occurs in the $\pm3/2\leftrightarrow\pm1/2$ transitions as is seen in the figures. The essential point is that the new eigen states $\Phi_{\pm1/2}$ are mixed states of nearly equal amounts of $|+1/2\rangle$ and $|-1/2\rangle$ together with small amounts of $|+3/2\rangle$ and $|-3/2\rangle$. Therefore the "forbidden transitions" have nearly equal intensities as the allowed transitions do, bringing the lowest doublet in Fig. 8(a) into two doublets like in Figs. 8(b) and 8(c). This feature is just seen in the experimental spectrum in Fig. 3(a).

On the other hand, the effect of the perpendicular field to the energetically higher levels is only of higher order and the $\pm7/2\leftrightarrow\pm5/2$ transitions change little with field from the zero-field NQR spectrum, keeping a single peak at low field region. However, the doublet at the highest frequency in Fig. 3(a) is easily produced if the magnetic field has a small Z-component by assuming that θ is deviated by a small angle from 90° as an example is shown in Fig. 8(d). The difference in Figs. 8(b) and 8(c) is in a difference of mixing between the states with $\Delta m=\pm2$ due to non-zero η term.

The low frequency branches in Fig. 8 which start from the origin are transitions between degenerate NQR states, $\Phi_{+1/2}\leftrightarrow\Phi_{-1/2}$ (solid curve) and $\Phi_{+3/2}\leftrightarrow\Phi_{-3/2}$ (broken curve, important for $\eta\gtrsim0.15$). A technique to extract information on the asymmetry parameter η free from magnitude and distribution of EFG by examining a powder pattern of the transitions between degenerate NQR states in a swept external field was discussed in cases of ^{75}As ($I=3/2$) in As_2S_3[14] and ^{181}Ta ($I=7/2$) in TaS_2[15] independently. The peak at 2.38 MHz in Fig. 3(a) seems to be the first example which is assigned as the transition between degenerate NQR states $\Phi_{\pm1/2}$ separated by an *internal field*.

After some calculations by trial and error, a set of parameters

$$\nu_Q=6.38\pm0.02\ \text{MHz},$$
$$\eta=0.01\pm0.01,$$
$$f_r=0.094\pm0.001\ (\text{or } H_N=0.997\pm0.010\ \text{kOe}),$$
$$\theta=78.0°\pm0.5°, \tag{6}$$

are found to reproduce the peak positions of

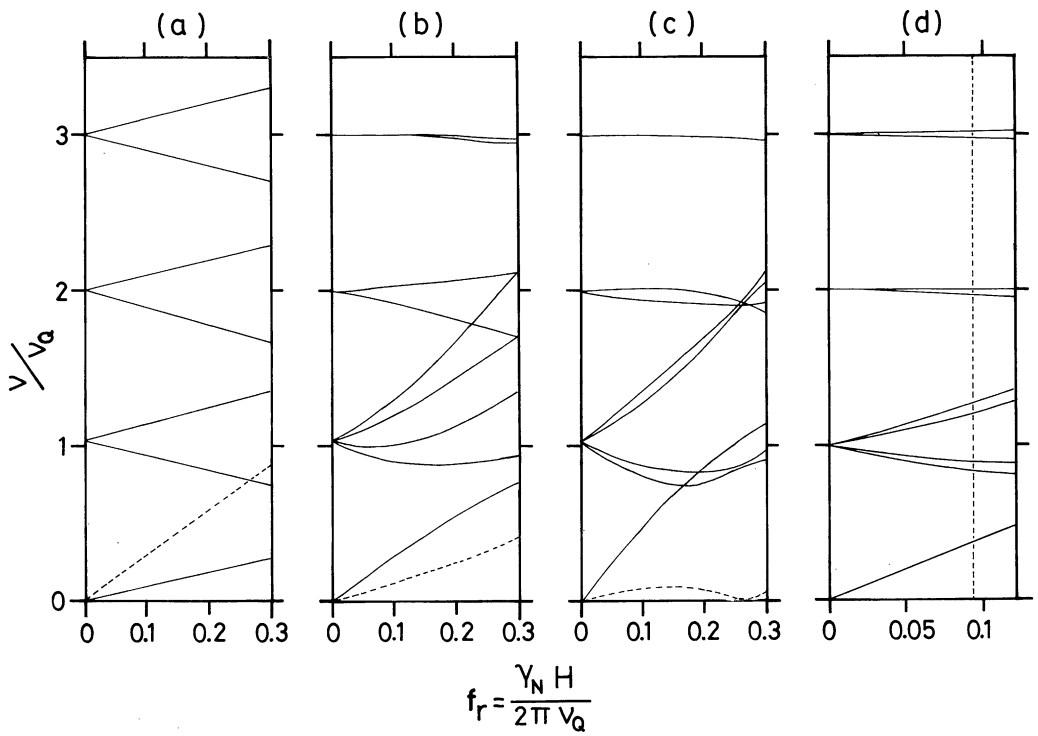


Fig. 8. Examples of calculated resonance frequencies in a unit of $\nu_Q(=3e^2qQ/2I(2I-1)h)$ as a function of a reduced strength of magnetic field, $f_r=\gamma_N H/2\pi\nu_Q$ are shown for the cases (a) $H/\!/Z$, (b) $H/\!/X$ and (c) $H/\!/Y$ for $\eta=0.1$. The case for a set of parameters: $\eta=0.01$, $\theta=78°$ and $\phi=0°$ is shown in (d) where dotted line shows the fitted case with $f_r=0.094$ (see text).

the experimental spectrum in Fig. 3(a) quite well and shown in Fig. 3(b) (see also Fig. 8(d)). The value of η agrees with that determined from the NQR spectrum at 293 K. Unfortunately, however, the value of ϕ could not be determined because of the small value of η. The intensity of the each transition has also been calculated from the transition probability $W(\theta_1, \phi_1)$ to the perturbation of the radio-frequency field $\boldsymbol{H}_1\ (\theta_1, \phi_1)\cdot\boldsymbol{I}$, where θ_1 and ϕ_1 are polar angles of $\boldsymbol{H}_1$ with respect to the principal axes. The intensity is shown in the figure as a length of each bar after taking powder average and a correction against frequency due to the Boltzmann factor. The agreement of the intensities between the experimental spectrum and the calculated one is qualitative. This is probably due to some inhomogeneous distribution of the internal field at the ^{139}La sites because, as are seen in Fig. 8(d), the resonance frequencies of $\pm 3/2 \leftrightarrow \Phi_{+1/2}$ and $\Phi_{+1/2} \leftrightarrow \Phi_{-1/2}$ transitions are rapidly varying functions of the field around the fitted value of f_r compared with other transitions and a little distribution of the hyperfine field brings severe broadening to the above two transitions.

Since the observed overall spectrum indicates that the distribution of the internal field is not so large, the magnetic structure of Cu spins must be simple. The incommensurate SDW state could be excluded. The reproduction of the spectrum with the small value of η seems to indicate that a coexistence of CDW is also less probable because conduction electrons in a CDW state often produce the field gradient at nuclear site strongly deviated from the axial symmetry as was observed in the case of NMR of ^{181}Ta in polytypes of TaS_2 and $TaSe_2$.[15]

The experimental result that the intensities of the doublet at 8.08/7.73 MHz and a single peak at 2.38 MHz for the sample prepared by the wet method are weaker than those for a sample prepared by the dry method seems to indicate that the former sample has somewhat more inhomogeneous distribution of the inter-

nal field on the contrary to the expectation. We suggest that microscopic sample quality could be checked by observing these transitions.

3.2 NMR spectrum

Although a sharp pattern in the spin-echo spectrum of ^{139}La in a swept field in Fig. 6 looks like an overlapped spectrum from two sites, the fields for every singular point of the pattern are shown to be consistent with those calculated from the same model with single site as was used for the NQR analysis. The numerical calculations using the same parameters as eq. (6) reproduce the every singular point as a result of the powder spectrum split by the internal field of 0.997 kOe nearly parallel and antiparallel to the applied field. The deviation of the powder spectrum from the ideally random distribution may be due to a preferred orientation of tiny crystals. The well articulated powder spectrum in Fig. 6 also suggests a simple spin structure and absence of CDW.

§4. Possible Spin Structure of Cu at Low Temperature Antiferromagnetic State

On the basis of the parameters (eq. (6)) to describe the NQR spectrum of ^{139}La in $La_2CuO_{4-\delta}$ at low temperatures, possible spin structures are discussed in this section. Although conflicting reports of the structure of La_2CuO_4 exist[4,5] we use the structure in ref. 5 for the present calculation since it was confirmed by powder neutron diffraction data.[6] A La atom is surrounded by nine oxygen atoms as shown in Fig. 9. EFG at a La site is calculated using the point-charge model which assumes the ionic states of O^{2-}, La^{3+} and Cu^{2+} to be

$$V_{ZZ}=eq=-8.5\times10^{13}\ \text{e.s.u.}, \tag{7a}$$

with

$$\eta=0.089, \tag{7b}$$

where all atoms within a distance of 40 Å are taken into account. The principal Z-axis (and the X-axis also) is found to be tilted by

$$\theta_Z=7.1°, \tag{7c}$$

within the ac-plane, while the principal Y-axis

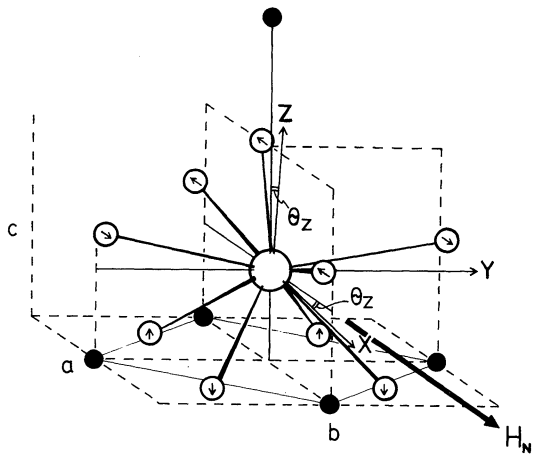


Fig. 9. Local arrangement of oxygen atoms around a lanthanum atom. The principal axes X, Y and Z of EFG calculated by the point charge model and the direction of the internal field H_N at the lanthanum site are shown (see text). Arrows show the directions of the displacements of the oxygen atoms.

with the intermediate field gradient coincides with the b-axis (b: minimum lattice constant) as shown in Fig. 9. Nine near neighbor oxygen atoms give dominant contributions to eq. (7) with $V_{ZZ}=-9.2\times10^{13}$ e.s.u., $\eta=0.089$ and $\theta_Z=5.9°$. The direction of the internal field $\boldsymbol{H}_N$ at a La site is tilted by 78° from the Z-axis and it does not coincide with the crystallographic axes in general. However, if we assume the point-charge calculation is correct only qualitatively, i.e., if we assume $\theta_Z=12°$ instead of eq. (7c), the direction of $\boldsymbol{H}_N$ includes the direction parallel to the a-axis. Hence, the most reasonable spin structure of Cu which can be deduced from the NQR data is a simple antiferromagnet with the sublattice moment along the a-axis.

A sum of classical dipole fields at a La site from Cu spins within a distance of 40 Å is calculated and shown in Table I for simple antiferromagnetic spin structures assuming each Cu spin has a moment of $1\mu_B$. As the internal field at a La site $\boldsymbol{H}_N$ is suggested here to be parallel to the a-axis, the types of the antiferromagnetic spin structures in the table which are consistent with the present analysis are those of 1, 4 or 7.

The present result should be compared with very recent experiments of neutron diffraction,[16,17] which suggest the antiferromagnetic

Table I. A sum of classical dipole fields in Oe at a lanthanum site from Cu spins within a distance of 40 Å for simple antiferromagnetic spin structures when each Cu spin has a moment of $1\mu_B$. The spin structure of each type is shown with a direction of the sublattice moment and $+/-$ sign at $A_1 \sim A_4$ sites of Cu atoms in Fig. 1.

Type	A_1 A_2 A_3 A_4	Direction of moment	$H_d(H_{dx}, H_{dy}, H_{dz})$
1		x	925 (−924, 0, −36)
2	+ − + −	y	991 (0, 991, 0)
3		z	76 (36, 0, 67)
4		x	393 (−393, 0, 17)
5	+ + − −	y	406 (0, 406, 0)
6		z	798 (−17, 0, 798)
7		x	998 (997, 0, 32)
8	+ − − +	y	920 (0, −920, 0)
9		z	84 (−32, 0, 77)

order of type 1 in Table I. An ordered sublattice moment (M_Q) of Cu is reported to be $0.5\pm0.15\mu_B$ for samples with T_N=220–240 K, although it depends strongly on oxygen treatment of samples: $M_Q=0.17\pm0.05$, 0.23 ± 0.05 and $0.43\pm0.13\mu_B$ for samples with T_N=50, 150 and 290 K, respectively.[16,17] The results of the neutron diffraction and the present NQR analysis are consistent with respect to the direction of the sublattice moments. A moment of about $0.5\mu_B$ determined by neutron diffraction indicates that a classical dipole field at a La site is about 0.5 kOe in the maximum from Table I. In order to match both the results, it may be considered that there must be a covalency between a La atom and near-neighbor Cu atoms in oblique directions to produce a transferred hyperfine field of ~0.5 kOe.

The NMR spectrum of ^{139}La in a strong external field at 4.2 K in Fig. 6 indicates that the spin-flop field (H_{SF}) is larger than the applied field of 80 kOe, because the signals at pairs of singular points indicated in Fig. 6 come from the La sites whose internal fields are nearly parallel and antiparallel to the applied field. This large value of H_{SF} seems to be quite unusual in copper compounds with this structure, since anisotropy of the dipole fields within the *ab*-plane at a copper site is only 3 ~ 18 Oe for simple antiferromagnetic structures in Table I, and spin-orbit coupling or anisotropic exchange interactions seems to be absent.

§5. Temperature Dependence of ν_Q and Sublattice Magnetization

Two physical quantities can be extracted from the temperature dependence of the highest double peaks in the NQR spectrum shown in Fig. 5. One is the temperature dependence of ν_Q which is obtained from averaged value of the upper and lower peaks and the other is the temperature dependence of the frequency difference between two peak positions which is related closely to that of the sublattice magnetization. The results of those are shown in Figs. 10(a) and 10(b), respectively.

The value of ν_Q decreases smoothly with increasing temperature that is probably associated with the temperature dependence of the unharmonic lattice vibration. It should be

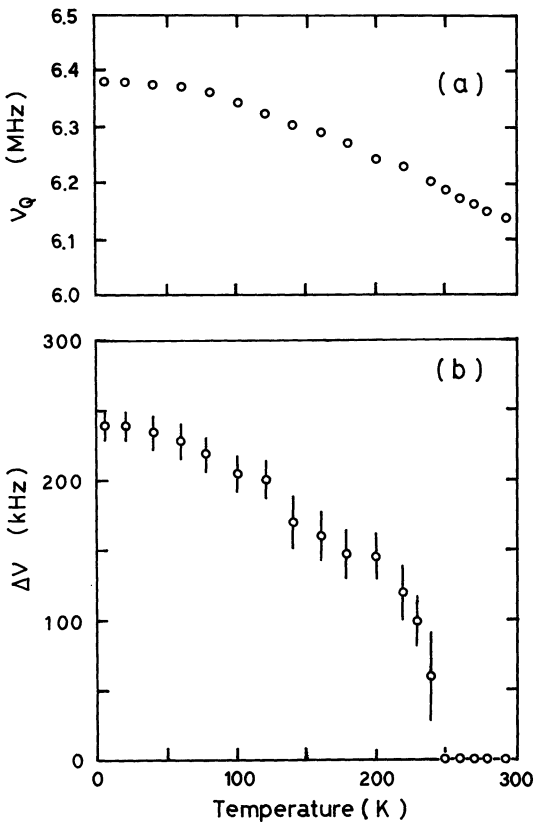


Fig. 10. (a) Temperature dependence of ν_Q at ^{139}La site in $La_2CuO_{4-\delta}$ and (b) that of the separation between the double peaks for $\pm7/2 \leftrightarrow \pm5/2$ transitions.

noted that no change in the crystallographical symmetry which may result in a discontinuous change in the field gradient was observed in the temperature range investigated.

The frequency separation of the double peaks in Fig. 3(a) has been shown to be caused by the internal field, however, the separation is not necessarily proportional to the internal field as is clearly seen in Fig. 8 where the frequency itself is not a linear function of f_r in general. A relation is shown in Fig. 11 between a separation $\Delta\nu$ and a reduced field f_r for the four doublets in a case of $\eta=0.01$ and $\theta=78°$. Fortunately, the relation is almost linear for the $\pm7/2\leftrightarrow\pm5/2$ transitions while small deviations exist for other transitions. Hence, the separation of the double peaks for $\pm7/2\leftrightarrow\pm5/2$ transitions shown in Fig. 10(b) is considered to be proportional to the internal field at the ^{139}La site, and to the Cu sublattice magnetization (M_Q). Therefore, $\Delta\nu(T)$ would be proportional to $M_Q(T)$, if we assume no change in θ_Z with temperature, and the fractional changes of $M_Q(T)$ in the reduced temperature (T/T_N with $T_N=250$ K) are shown in Fig. 12. It is immediately seen that $M_Q(T)/M_Q(0)$ decreases much faster than that expected from a molecular field approximation with well defined local moment of $S=1/2$. The solid curve in Fig. 12 indicates the theoretical temperature dependence based on the spin fluctuation theory for weak itinerant antiferromagnets which is expressed as follows;[18]

$$M_Q(T)/M_Q(0)=[1-(T/T_N)^{3/2}]^{1/2}. \quad (8)$$

The agreement between the theory and the experiment is satisfactory although some deviations are observed near T_N. This fact together with the fact that the uniform susceptibility has essentially no temperature dependence compared with the local moment case enforced us to believe that the wave vector q-dependent spin fluctuations with only components around $q=Q$ (Q being antiferromagnetic wave vector) play an important role to the magnetic properties. Here, one should recall the same argument made for the typical weakly itinerant antiferromagnet V_3Se_4, where the same behaviors for $M_Q(T)$ and χ have been observed and discussed.[19]

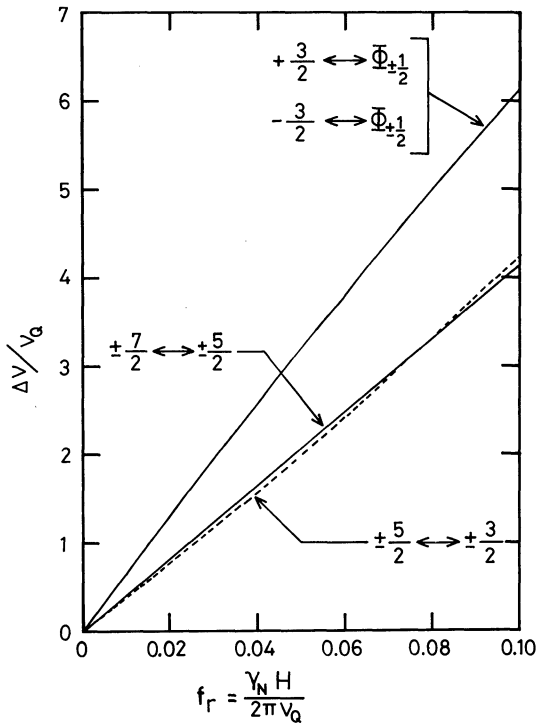


Fig. 11. Calculated frequency separation normalized to ν_Q of the four double peaks in Fig. 3(b) versus reduced magnetic field $f_r=\gamma_N H/2\pi\nu_Q$ in a case of $\eta=0.01$, $\theta=78°$ and $\phi=0°$.

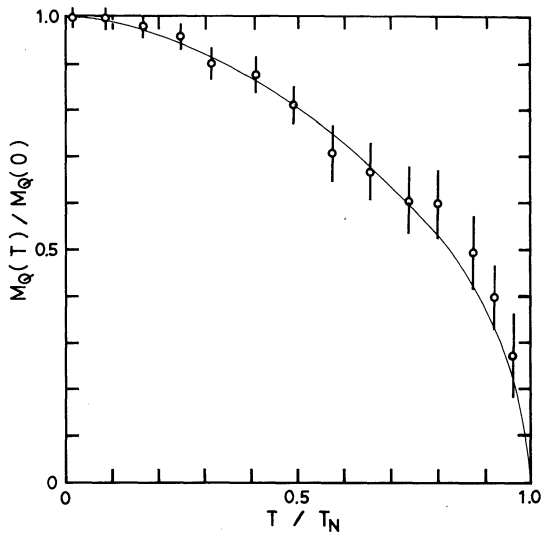


Fig. 12. Temperature dependence of the normalized sublattice magnetization deduced from Fig. 10(b). The solid curve shows a theoretical prediction for weakly itinerant antiferromagnet (see text).

§6. Concluding Remarks

The NQR and NMR studies of ^{139}La in $La_2CuO_{4-\delta}$ reported here establish the basic properties of the antiferromagnetism. It is well known that the antiferromagnetism, both of the T_N and M_Q, strongly depend on the value of δ. Although we could not extract the absolute value of M_Q from the present ^{139}La NQR or NMR experiments, T_N of our sample is determined to be 250 ± 2 K. This rather high T_N indicates that our sample is in an oxygen deficient phase, yielding a distribution of the electric field gradient to some extent that is actually observed in the spectrum at room temperature. The important results emerged from the present investigation are summarized below.

(i) Rather complex NQR spectrum in the antiferromagnetic state ($T=1.3$ K) is entirely interpreted by solving a secular equation for the Hamiltonian which includes both the quadrupole and Zeeman interactions. The quadrupole coupling constant and the asymmetry parameter were obtained to be $\nu_Q=6.38\pm0.02$ MHz and $\eta=0.01\pm0.01$, respectively. The internal field acting on the La sites is also determined to be 0.997 ± 0010 kOe which is nearly perpendicular to the direction of the maximum field gradient.

(ii) Combined above results with the calculations of the field gradient using the point charge model and the classical dipole field from the Cu moments, a simple antiferromagnetic spin structure where the direction of Cu moment is either parallel or antiparallel to the a-axis is proposed. This agrees with the recent neutron diffraction experiments. However, the value of moment estimated from simple dipole field calculation is about twice as large as that obtained by the neutron experiments. The disagreement may come from a sizable contribution of the transferred hyperfine field to the internal field at the La sites.

(iii) The NMR spectrum in a strong external field at 4.2 K suggests that the spin-flop field is larger than the maximum applied field of 80 kOe.

(iv) From the smooth temperature variation of ν_Q, it is concluded that there is no evidence for any structural phase transition between 1.3 K and room temperature.

(v) The temperature dependence of the internal field which is proportional to the sublattice magnetization of the Cu atoms has been obtained. The fractional change of the magnetization with temperature is fitted to the theoretical prediction based on the spin fluctuations in weakly itinerant antiferromagnets. The obtained agreement between the theory and the experiment suggests that the spin fluctuations with only components around $q=Q$ play an important role to the magnetic properties, rather than the local moment fluctuations.

In this paper, we have restricted ourselves to report only the static part of the NQR and NMR results. Needless to say, the dynamical studies are of particular importance to understand the magnetic excitations including the critical behaviors near T_N. Those can be done through the measurement of nuclear relaxation time with varying temperature and will be the subject of our next paper.

Notes added in a draft: After this work was completed and when this draft was almost ready, four preprints of letter articles come to our notice.[20-23] Zero-field NQR spectrum is analyzed with a first-order[21,23] or second-order perturbation theory[22] in contrast to the present numerical analysis with similar parameters. However, the conclusion that the spin structure of Cu is complex obtained by neglecting the orthorhombic distortion of the crystal[21,22] seems to be unlikely.

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